



Lect. 5 Spatial Correlation Analysis

□ Traditional Correlation Analysis in Geography

1. Person's Correlation Coefficient

2. Spearman's Rank Correlation Coefficient
3. Partial Correlation Coefficient
4. Multiple Correlation Coefficient

□ Spatial Autocorrelation Analysis

1. The Concept of Spatial Autocorrelation
2. Spatial Weights Matrix
3. Global Spatial Autocorrelation
4. Local Indicators of Spatial Association

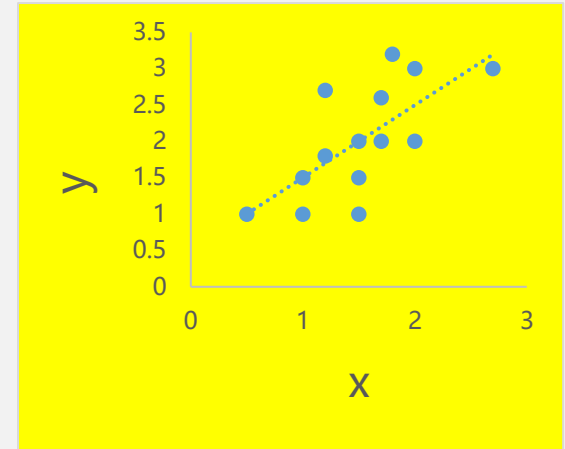
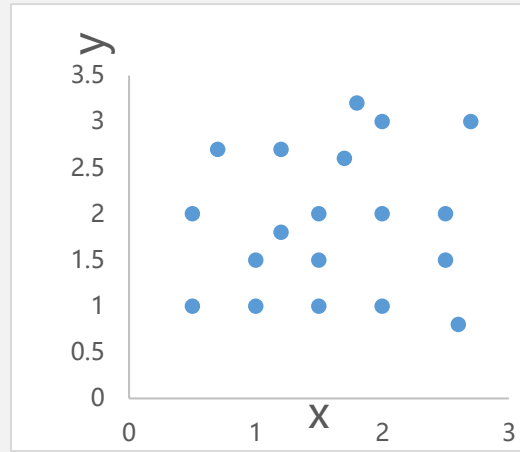
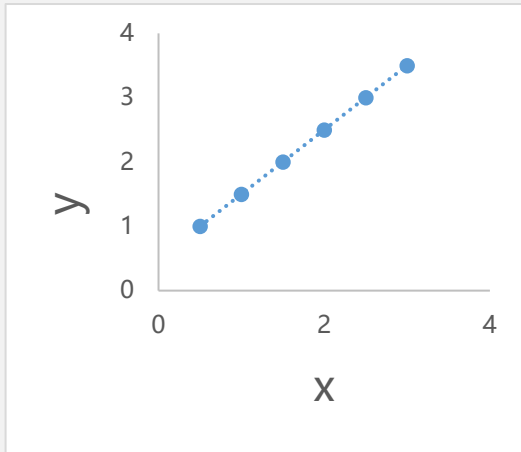
□ Study Cases



1. Person's Correlation Coefficient

Geographical correlation types:

- Complete correlation
- No correlation
- Statistical correlation



1. Person's Correlation Coefficient

Pearson's correlation coefficient:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$



1. Person's Correlation Coefficient

➤ The sample correlation coefficient:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n - 1)s_x s_y}$$

Pearson's correlation coefficient

Where s_x and s_y are the sample standard deviations of variables x and y , respectively

Note that r is also equal to

$$r = \frac{\sum_{i=1}^n Z_x Z_y}{(n - 1)}$$

Where z_x and z_y are z-scores associated with variables x and y , respectively

1. Person's Correlation Coefficient

➤ The population correlation coefficient:

$$\rho = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}$$

- μ_x and μ_y are the mean values of variables x and y in the population
- σ_x and σ_y refer to the standard deviations of variables x and y

- $\text{Cov}[X, Y]$ is the Covariance (协方差) of variables x and y

Covariance is a straightforward extension of the concept of variance. Whereas the variance is the expected or average value of the squared deviation of observations on a single variable from their mean, the covariance is the expected or average value of the product of the two variables' (say X and Y) respective deviations from their means.

- Note that the covariance of a variable x with itself, $\text{Cov}(X, X)$, is equal to the variance of X .

Sample
covariance:

$$\text{Cov}[X, Y] = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

1. Person's Correlation Coefficient

➤ The population correlation coefficient:

$$\rho = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}$$

The magnitude of the covariance will depend upon the units of measurement.

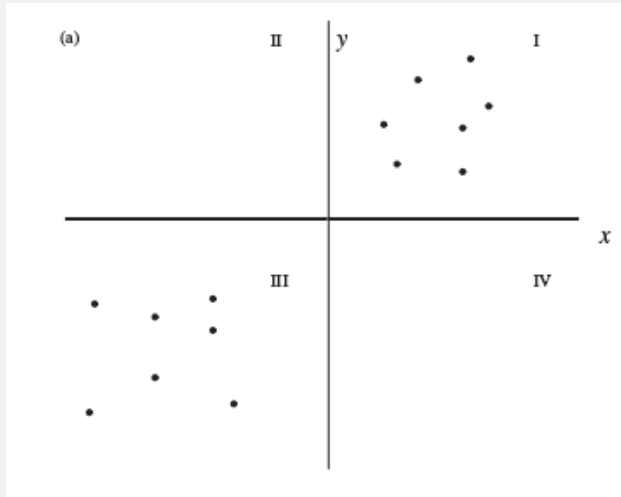
- The covariance may be standardized, so that its values lie in the range from -1 to +1, by dividing by the product of the standard deviations.
- This standardized covariance is known as **Pearson's correlation coefficient**.
- The correlation coefficient provides a standardized measure of the linear association between two variables.

1. Person's Correlation Coefficient

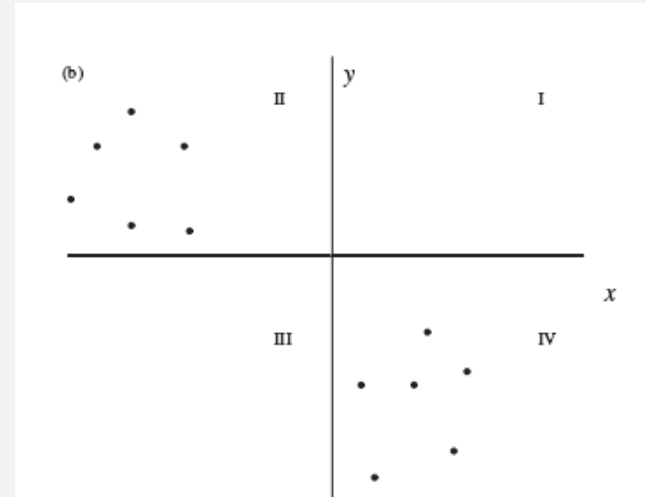
➤ The population correlation coefficient:

The covariance of X and Y may be **negative** or **positive**.

- The covariance will be positive if most of the points (x, y) lie along a line with positive slope when they are plotted.
- The covariance will be negative when the plotted points lie along a line with negative slope.



Negative



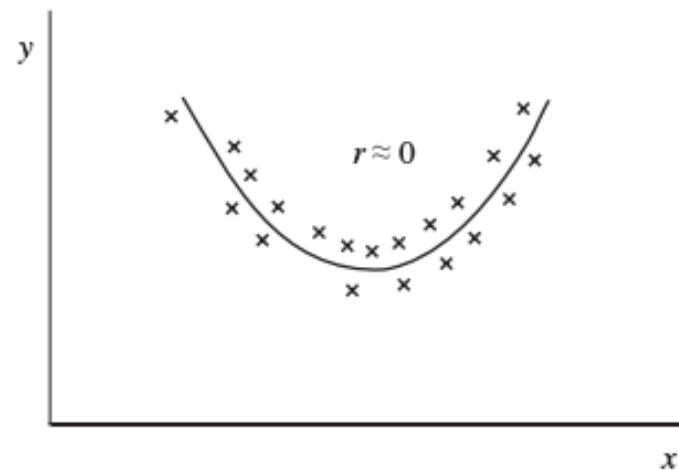
Positive

1. Person's Correlation Coefficient

$$r \in [-1,1]$$

It is important to note that the correlation coefficient is a measure of the strength of the linear association between variables.

- As the right figure demonstrates, it is possible to have a strong, nonlinear association between two variables and yet have a correlation coefficient close to zero.
- One implication of this is that it is important to plot data since potential associations between the variables might be revealed in those cases where the value of r is low.



1. Person's Correlation Coefficient

➤ Correlation Coefficient Matrix

$$R = \begin{bmatrix} r_{11} & \cdots & r_{1n} \\ \vdots & \ddots & \vdots \\ r_{n1} & \cdots & r_{nn} \end{bmatrix}$$

r_{ij} represents the correlation coefficient between two random variables

R is symmetrical matrix,
the catercorner elements' values are all equal to 1.



Illustrations of Correlation Coefficient

The existence of a strong linear association does not necessarily imply that there is a causal connection (因果关系) between the two variables.

- For example, strong correlation was once found between British coal production and the death rate of penguins (企鹅) in the Antarctic, but it would be a stretch of the imagination to connect the two in any direct way.
- Another article once pointed out the strong connection between the annual number of tornados and the volume of automobile traffic in the United States. The claim - presumably in jest - was that both the number of tornados and the volume of automobile traffic had steadily increased in the United States throughout the twentieth century.

Illustrations of Correlation Coefficient

Mobility and Cohort Size

Cohort refers to the baby boom)

For 25-29 year olds:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y} = -0.747$$

Table 5.1 U.S. mobility data, 1948-1984

Year	Mobility rate		Fraction of total population	
	20-24	25-29	20-24	25-29
1948-49	35.0	*	.0804	.0829
1949-50	34.0	*	.0784	.0821
1950-51	37.7	33.6	.0767	.0812
1951-52	37.8	31.6	.0746	.0794
1952-53	40.5	33.4	.0720	.0774
1953-54	38.1	30.5	.0757	.0762
1954-55	41.8	31.3	.0735	.0758
1955-56	44.5	32.3	.0713	.0750
1956-57	41.2	32.0	.0694	.0734
1957-58	42.6	34.6	.0671	.0715
1958-59	42.5	33.2	.0645	.0701
1959-60	41.2	32.1	.0617	.0682
1960-61	43.6	34.4	.0616	.0605
1961-62	43.2	33.0	.0625	.0592
1962-63	42.0	34.6	.0641	.0582
1963-64	43.4	35.2	.0672	.0580
1964-65	45.0	35.8	.0692	.0528
1965-66	42.4	35.5	.0708	.0584
1966-67	41.0	33.0	.0715	.0593
1967-68	41.5	33.2	.0767	.0609
1968-69	42.5	32.6	.0787	.0638
1969-70	41.8	32.6	.0813	.0658
1970-71	41.2	32.4	.0839	.0669
1975-76	38.0	32.6	.0897	.0790
1980-81	36.8	30.1	.0943	.0859
1981-82	35.5	30.0	.0949	.0868
1982-83	33.7	29.8	.0939	.0892
1983-84	34.1	30.1	.0928	.0904

Note: Data unavailable for missing years.

Source: Rogerson (1987).

A Significance Test for r

To test the **null hypothesis that the true correlation coefficient, ρ , is equal to zero**, the data for each variable are assumed to come from normal distributions.

If this assumption is satisfied, the test may be carried out by forming the t -statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

If the null hypothesis is true, this statistic has a t -distribution, with $n-2$ degrees of freedom.



A Significance Test for r

Mobility and Cohort Size

- $n = 28$
- $r = -0.747$
- For the null hypothesis $H_0: \rho = 0$
 - with a two-tailed test with $\alpha = 0.05$
 - the t -statistic is

$$t = \frac{(-0.747)\sqrt{28-2}}{\sqrt{1-(-0.747)^2}} = -5.73$$

Since the critical values of t , using $\alpha = 0.05$ in a two-tailed test with 26 degrees of freedom are ± 2.056 , we **rejected the null hypothesis**

	P(2):双侧	0.5	0.2	0.1	0.05	0.02	0.01
n'	P(1):单侧	0.25	0.1	0.05	0.025	0.01	0.005
1		1	3.078	6.314	12.706	31.821	63.657
2		0.816	1.886	2.92	4.303	6.965	9.925
3		0.765	1.638	2.353	3.182	4.541	5.841
4		0.741	1.533	2.132	2.776	3.747	4.604
5		0.727	1.476	2.015	2.571	3.365	4.032
6		0.718	1.44	1.943	2.447	3.143	3.707
7		0.711	1.415	1.895	2.365	2.998	3.499
8		0.706	1.397	1.86	2.306	2.896	3.355
9		0.703	1.383	1.833	2.262	2.821	3.25
10		0.7	1.372	1.812	2.228	2.764	3.169
11		0.697	1.363	1.796	2.201	2.718	3.106
12		0.695	1.356	1.782	2.179	2.681	3.055
13		0.694	1.35	1.771	2.16	2.65	3.012
14		0.692	1.345	1.761	2.145	2.624	2.977
15		0.691	1.341	1.753	2.131	2.602	2.947
16		0.69	1.337	1.746	2.12	2.583	2.921
17		0.689	1.333	1.74	2.11	2.567	2.898
18		0.688	1.33	1.734	2.101	2.552	2.878
19		0.688	1.328	1.729	2.093	2.539	2.861
20		0.687	1.325	1.725	2.086	2.528	2.845
21		0.686	1.323	1.721	2.08	2.518	2.831
22		0.686	1.321	1.717	2.074	2.508	2.819
23		0.685	1.319	1.714	2.069	2.5	2.807
24		0.685	1.318	1.711	2.064	2.492	2.797
25		0.684	1.316	1.708	2.06	2.485	2.787
26		0.684	1.315	1.706	2.056	2.479	2.779
27		0.684	1.314	1.703	2.052	2.473	2.771
28		0.683	1.313	1.701	2.048	2.467	2.763
29		0.683	1.311	1.699	2.045	2.462	2.756
30		0.683	1.31	1.697	2.042	2.457	2.75

The Correlation Coefficient and Sample Size

- An extremely important point : the correlation coefficient is influenced by sample size.
- It is far easier to reject the null hypothesis that $\rho = 0$ with a large sample size than it is with a small sample size.
- To see this, compare the situation where $r = 0.4$ with a sample size of 11, and the situation where $r = 0.4$ with a sample size of 38.
 - In the former case, the observed t -statistic is 1.3, which is less than the critical value $t_{0.05,9} = 2.262$, and the null hypothesis is accepted.
 - In the latter case, when $n = 38$, the t -statistic is 2.0, and the null hypothesis is rejected since the t -statistic is greater than the critical value, $t_{0.05,36} = 1.96$.



The Correlation Coefficient and Sample Size

- An extremely important point : the correlation coefficient is influenced by sample size.

Table 5.2 Minimum values of r required for significance

Sample size, n	Minimum absolute value of r needed to attain significance (using $\alpha = 0.05$)
15	.514
20	.444
30	.361
50	.279
100	.197
250	.124

For large n , r_{crit} is approximately $2/\sqrt{n}$.

- It implies that there really should be no popular rules of thumb that are invoked to decide whether r is sufficiently high to make the researcher happy about the level of correlation.



The Correlation Coefficient and Sample Size

- Such rules of thumb do seem to exist -- *for instance, an r value of 0.7 or 0.8 may be taken as important or significant.*
- Whether a correlation coefficient is truly significant depends upon the sample size.
- Thus, when data set is large, a relatively low value of r should not be as disappointing as that same value of r when the sample size is smaller.
- A value of r 0.4 could be quite meaningful if n is 1000, and we should not necessarily throw the results out the window just because the r -value is noticeably less than 1 and perhaps less than some arbitrary, rule-of-thumb value such as r 0.8.
- Meehl (1990) has noted that with many data sets there is a strong tendency to find that “everything correlates to some extent with everything else” .
- There is no particular reason for believing that the correlation between any two variables chosen from most data sets should be exactly zero, and therefore, if the sample size is large enough, we will be able to reject the null hypothesis that they are unrelated.

2. Spearman's Rank Correlation Coefficient

- In the situations where only ranked data are available, or where the assumption of normality required for the test $H_0: \rho = 0$ is not satisfied, it is appropriate to use **Spearman's rank correlation coefficient** - 秩 (等级) 相关系数.
- Two separate sets of ranks are developed, one for each variable. A rank of 1 is assigned to the lowest value and a rank of n to the highest observation in each column.

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n^3 - n}$$

Where d_i^2 is the squared difference between the ranks for observation i , and n is the sample size.



2. Spearman's Rank Correlation Coefficient

The mobility example

$$r_s = -0.725$$

- To test hypothesis, we may use the fact that $r_s \sqrt{n-1}$ has a t -distribution with $n-1$ degrees of freedom.
- The observed value of t is -3.70, it is less than the critical value $t_{0.05,25} = -2.06$, so we rejected the null hypothesis

Table 5.3 U.S. mobility data, 1948–84, with ranks

Year	Mobility rate		Fraction of total population		d_i
	20–24	Rank	20–24	Rank	
1950–51	37.7	5	.0767	17.5	-12.5
1951–52	37.8	6	.0746	15	-9
1952–53	40.5	9	.0720	13	-4
1953–54	38.1	8	.0757	16	-8
1954–55	41.8	15.5	.0735	14	1.5
1955–56	44.5	25	.0713	11	14
1956–57	41.2	12	.0694	9	3
1957–58	42.6	21	.0671	6	15
1958–59	42.5	19.5	.0645	5	14.5
1959–60	41.2	12	.0617	2	10
1960–61	43.6	24	.0616	1	23
1961–62	43.2	22	.0625	3	19
1962–63	42.0	17	.0641	4	13
1963–64	43.4	23	.0672	7	16
1964–65	45.0	26	.0692	8	18
1965–66	42.4	18	.0708	10	8
1966–67	41.0	10	.0715	12	-2
1967–68	41.5	14	.0767	17.5	-3.5
1968–69	42.5	19.5	.0787	19	0.5
1969–70	41.8	15.5	.0813	20	-4.5
1970–71	41.2	12	.0839	21	-9
1975–76	38.0	7	.0897	22	-15
1980–81	36.8	4	.0943	25	-21
1981–82	35.5	3	.0949	26	-23
1982–83	33.7	1	.0939	24	-23
1983–84	34.1	2	.0928	23	-21

Source: Plane and Rogerson (1991).

The effect of spatial dependence on significance tests for correlation coefficient

- The tests of significance outlined for both **Pearson's r** and **Spearman's r_s** assume that the observations of x and y are **independent**.
- When the x and y variables come from spatial locations, this assumption of independence may not be satisfied. Indeed, one of the most important points in our course is that **spatial data often exhibit dependence** – the value of x in one location is often related to the value of x in nearby locations.
- In turn, spatial dependence affect the outcome of statistical tests, and we should always keep this point in mind when interpreting statistical results.

Illustrations

The effect of spatial dependence on significance tests for correlation coefficient.

- The critical values associated with the standard statistical tests (b and c) are lower in absolute value than those which should be used (a and d).
- When sample values of r fall in the shaded region, the standard statistical tests will incorrectly imply that the correlation coefficient is significantly different from zero.
- Correlation coefficient falling in the shaded region are likely not significant, and may be the result of the underlying spatial dependence exhibited by the x and y variables.

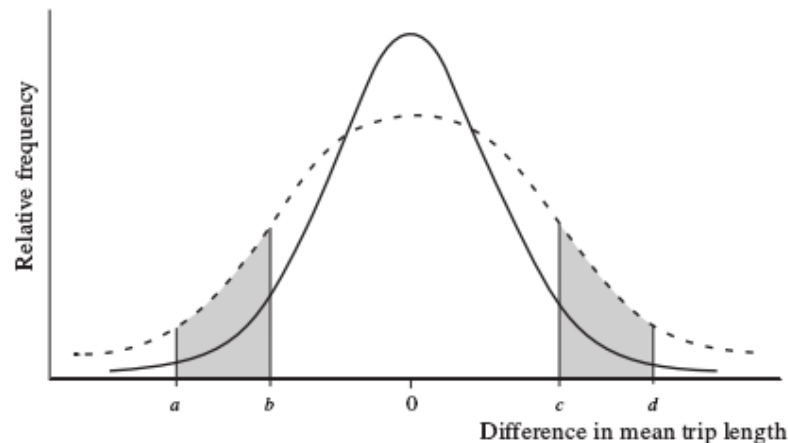


Figure 5.5 Assumed (—) and actual (---) variability in r when spatial dependence is present but not accounted for

3. Partial Correlation Coefficient

- **Partial correlation coefficient** is used to measure the degree of association between two random variables, with the effect of **a set of controlling random variables removed**.
- If we are interested in finding whether or to what extent there is a numerical relationship between two variables of interest, using their correlation coefficient will give misleading results if there is another variable that is numerically related to both variables of interest.
- This misleading information can be avoided by controlling for the confounding variable, which is done by computing the partial correlation coefficient.
- This is precisely the motivation for including other right-side variables in a multiple regression.

3. Partial Correlation Coefficient

Example : 4 variables, 24 cases

Case#	x_1	x_2	x_3	x_4
1	679	342.9	141.8	194.3
2	824	378	192.5	253.5
3	859	392	211.7	252.2
4	910	421	222.2	266.8
. . .	. . .	. . .	. . .	. . .
24	81910.9	14457.2	40417.9	27035.8

3. Partial Correlation Coefficient

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

$$r_{xy.z} = \frac{r_{xy} - r_{xz}r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}}$$

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

$$r_{13.2} = \frac{r_{13} - r_{12}r_{23}}{\sqrt{(1 - r_{12}^2)(1 - r_{23}^2)}}$$

$$r_{23.1} = \frac{r_{23} - r_{12}r_{13}}{\sqrt{(1 - r_{12}^2)(1 - r_{13}^2)}}$$

3. Partial Correlation Coefficient

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ r_{41} & r_{42} & r_{43} & r_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0.416 & 0.346 & 0.579 \\ 0.416 & 1 & -0.592 & 0.950 \\ 0.346 & -0.592 & 1 & -0.469 \\ 0.579 & 0.950 & -0.469 & 1 \end{bmatrix}$$

$r_{12 \cdot 3}$	$r_{13 \cdot 2}$	$r_{14 \cdot 2}$	$r_{14 \cdot 3}$	$r_{23 \cdot 1}$	$r_{24 \cdot 1}$	$r_{24 \cdot 3}$	$r_{24 \cdot 1}$	$r_{34 \cdot 2}$
0.821	0.808	0.647	0.895	-0.863	0.956	0.945	-0.875	0.371

$$r_{12 \cdot 34} = \frac{r_{12 \cdot 3} - r_{14 \cdot 3} r_{24 \cdot 3}}{\sqrt{(1 - r_{14 \cdot 3}^2)(1 - r_{24 \cdot 3}^2)}} = \frac{0.821 - 0.895 \times 0.945}{\sqrt{(1 - 0.895^2)(1 - 0.945^2)}} = -0.170$$

$r_{12 \cdot 34}$	$r_{13 \cdot 24}$	$r_{14 \cdot 23}$	$r_{23 \cdot 14}$	$r_{24 \cdot 13}$	$r_{34 \cdot 12}$
-0.170	0.802	0.635	-0.187	0.821	-0.337

Refer to “郑培根 刘梓修. 偏相关系数的定义及计算[J]. 江西财经学院学报, 1986(01):79-83” get more information about partial correlation.

3. Partial Correlation Coefficient

当有 m 个变量，设变量集合为 $X_1, X_2, \dots, X_m$,

要计算在固定其他 $m - 2$ 个变量时，即控制特定的一个子集 S ，变量 X_i 与 X_j 之间的偏相关系数可以通过递归关系从低阶偏相关系数

$$\rho_{ij \cdot S} = \frac{\rho_{ij \cdot S \setminus \{t\}} - \rho_{it \cdot S \setminus \{t\}} \rho_{jt \cdot S \setminus \{t\}}}{\sqrt{1 - \rho_{it \cdot S \setminus \{t\}}^2} \sqrt{1 - \rho_{jt \cdot S \setminus \{t\}}^2}}$$

其中 t 是集合 S 中的任意一个变量， $S \setminus \{t\}$ 表示从控制集中去掉 t 。
当 S 包含所有 $m - 2$ 个其他变量时，即得所需的偏相关系数。

3. Partial Correlation Coefficient

Characteristic: $r_{xy \cdot z} \in [-1, 1]$

Significance test:

$$t = \frac{r_{12 \cdot 34 \dots m}}{\sqrt{1 - r_{12 \cdot 34 \dots m}^2}} \sqrt{n - m - 2}$$

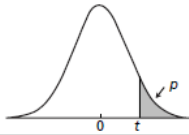
$r_{12 \cdot 34}$	$r_{13 \cdot 24}$	$r_{14 \cdot 23}$	$r_{23 \cdot 14}$	$r_{24 \cdot 13}$	$r_{34 \cdot 12}$
-0.170	0.802	0.635	-0.187	0.821	-0.337

For $r_{24 \cdot 13} = 0.821$, $t = 22.432$

$t_{0.05, 19} = 1.729$

Table A.3 Student's *t* distribution

For various degrees of freedom (df), the tabled entries represent the critical values of *t* above which a specified proportion *p* of the *t* distribution falls. (Example: for df=9, a *t* of 2.262 is surpassed by .025 or 2.5% of the total distribution.)



df	<i>p</i> (one-tailed probabilities)				
	.10	.05	.025	.01	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Adapted from Table III of Fisher and Yates (1974).

4. Multiple Correlation Coefficient

- **The coefficient of multiple correlation** (复相关系数) is a measure of how well a given variable can be predicted using a linear function of a set of other variables.

$$R = \sqrt{r_{yx}^T R_{xx}^{-1} r_{yx}}$$

$$R \in [0,1]$$

其中 r_{yx} 为 $m \times 1$ 的列向量, 其中每个元素是 y 与每个自变量 x_i 的皮尔逊Person相关系数

R_{xx} 为 $m \times m$ 的自变量之间的相关系数矩阵

r_{yx}^T 是 r_{yx} 的转置

R_{xx}^{-1} 是自变量相关系数矩阵的逆矩阵

- **决定系数:** 表示在多元线性回归中, 自变量 $x_1, x_2, \dots, x_m$ 共同解释的因变量 y 的方差比例 (第6讲)

$$R^2 = r_{yx}^T R_{xx}^{-1} r_{yx} = \text{回归平方和} / \text{总平方和}$$

4. Multiple Correlation Coefficient--Test

Example: $R = 0.974$, $n=23$, $m=3$,

Sum of Squares
for Regression

The number of total
variables $k=m+1$

$$F = \frac{SSR/(k-1)}{SSE/(n-k-1)} = \frac{0.974^2}{1-0.974^2} \times \frac{23-3-1}{3} = 117.0657$$

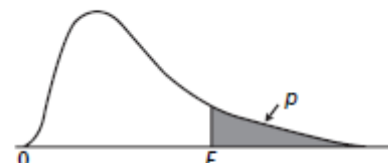
Sum of
Squares
for Error

$$F = 117.0657 > F_{0.01} = 5.01$$

$\therefore R$ is significant !

Table A.5 *F* distribution

For various pairs of degrees of freedom v_1, v_2 , the tabled entries represent the critical values of F above which a proportion p of the distribution falls. (Example: for $df=4, 16$ an $F=2.33$ is exceeded by $p \geq .10$ of the distribution.) Tables are provided for values of p equal to .10, .05, .01.

 **$p = .01$ Values**

Degrees of freedom for denominator	Degrees of freedom for numerator, v_1																
	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	∞
1	4052	4999.5	5403	5625	5764	5859	5928	5981	6022	6056	6106	6157	6209	6261	6287	6313	6366
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.47	99.47	99.48	99.50
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.50	26.41	26.32	26.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.84	13.75	13.65	13.56
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.38	9.29	9.20	9.11
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.23	7.14	7.06	6.97
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	5.99	5.91	5.82	5.74
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.20	5.12	5.03	4.95
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.65	4.57	4.48	4.40
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.25	4.17	4.08	4.00
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	3.94	3.86	3.78	3.69
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.70	3.62	3.54	3.45
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.51	3.43	3.34	3.25
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.35	3.27	3.18	3.09
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.21	3.13	3.05	2.96
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.10	3.02	2.93	2.84
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.00	2.92	2.83	2.75
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	2.92	2.84	2.75	2.66
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.84	2.76	2.67	2.58
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.78	2.69	2.61	2.52
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.72	2.64	2.55	2.46
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.67	2.58	2.50	2.41
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.62	2.54	2.45	2.35
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.58	2.49	2.40	2.31
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.39	2.30	2.21	2.11
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.20	2.11	2.02	1.92
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.03	1.94	1.84	1.73
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.86	1.76	1.66	1.53
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.70	1.59	1.47	1.32



Lect. 5 Spatial Correlation Analysis

□ Traditional Correlation Analysis in Geography

1. Person's Correlation Coefficient
2. Spearman's Rank Correlation Coefficient
3. Partial Correlation Coefficient
4. Multiple Correlation Coefficient

□ Spatial Autocorrelation Analysis

1. The Concept of Spatial Autocorrelation

2. Spatial Weights Matrix
3. Global Spatial Autocorrelation
4. Local Indicators of Spatial Association

□ Study Cases



1. The Concept of Spatial Autocorrelation

The first law of geography: Everything is related to everything else, but near things are more related than distant things-- *Waldo Tobler*.

- One of the most common objectives of researchers is to determine whether two variables are associated with one another.
 - For examples: Does interaction vary with distance? Do housing prices vary with accessibility to major highways?
 - Researchers are interested in how variables **covary**(共变) .
- This first law is the foundation of the fundamental concepts of **spatial dependence** and **spatial autocorrelation** (空间自相关) and is utilized specifically for the inverse distance weighting method for spatial interpolation and to support the regionalized variable theory for kriging.

1. The Concept of Spatial Autocorrelation

Spatial autocorrelation is the use of spatial statistics to study the spatial correlation between spatial units and their surrounding units, so as to analyze the spatial distribution characteristics of these spatial units.

- If the degree of spatial autocorrelation is high, the spatial phenomena with the same characteristics will gather together.
- On the contrary, if the degree of spatial autocorrelation is low, the spatial phenomena may be scattered throughout the space.

Spatial Correlation Analysis means applying the methods of correlation analysis to study spatial distributed data.

2. Spatial Weights Matrix

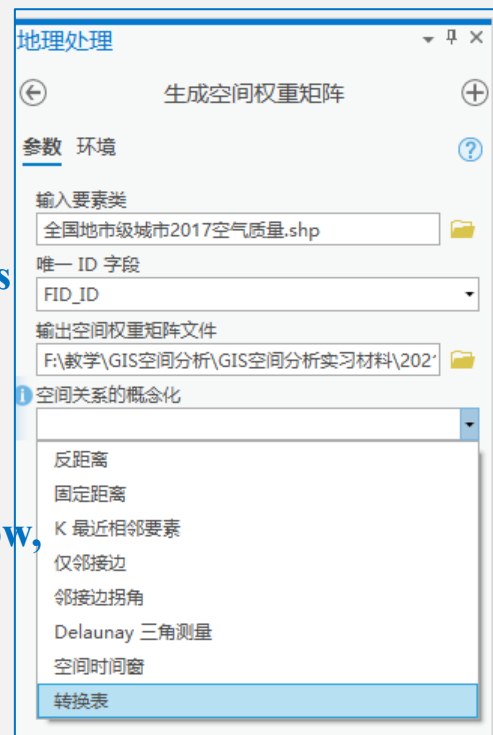
- Spatial statistics does not mean applying traditional (non-spatial) statistical methods to data that just happens to be spatial (having x - and y -coordinates).
- **Spatial statistics** integrate space and spatial relationships directly into their mathematics (area, distance, length, and so on). For many spatial statistics, these spatial relationships are specified formally through a spatial weights matrix file or table.
- A **spatial weights matrix** is a representation of the **spatial relationships** among features in a dataset. It is a quantification of the spatial relationships that exist among the features in a dataset (or, at least, a quantification of the way you conceptualize those relationships).
- Because **the spatial weights matrix imposes a structure** on your data, we should select **a conceptualization** that best reflects how features actually interact with each other (giving thought, of course, to what it is you are trying to measure).

2. Spatial Weights Matrix

The Conceptualization of Spatial Relationship



The conceptualization of Spatial relationships in ArcGIS Pro

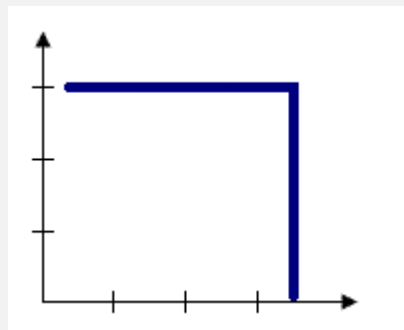


The **Convert table** option can be used to convert an [ASCII spatial weights matrix file](#) to a [SWM formatted spatial weights matrix file](#).

Spatial Weights Matrix: Distance

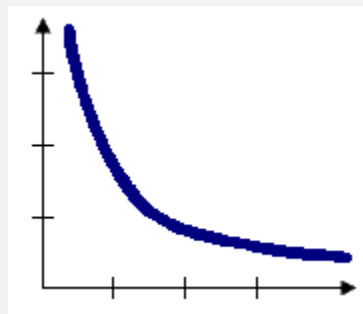
Threshold distance (Distance band):

$$w_{ij} = \begin{cases} 1, d < D \\ 0, otherwise \end{cases}, D \text{ is the determined distance.}$$



Inverse distance:

$$w_{ij} = \frac{1}{d_{ij}^2}$$



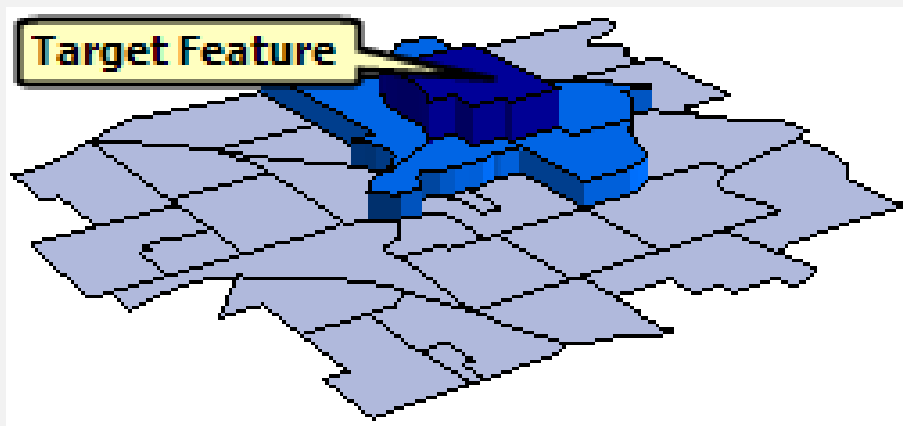
Spatial Weights Matrix: Polygon Continuity

$$W = \begin{bmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{bmatrix}$$

The construction of spatial weights matrix is generally based on two characteristics: **continuity and distance**.

It can also be constructed by area and accessibility.

Where w_{ij} indicates spatial weight index between regions i and j .



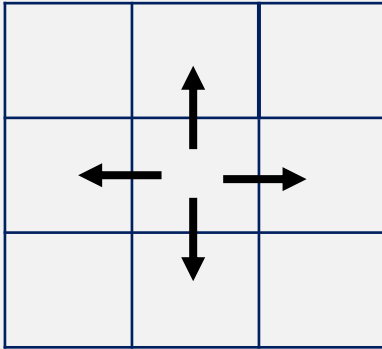
*Spatial relationships based on **polygon continuity**, Queen's case: shared edges or nodes*

Spatial Weights Matrix: Continuity

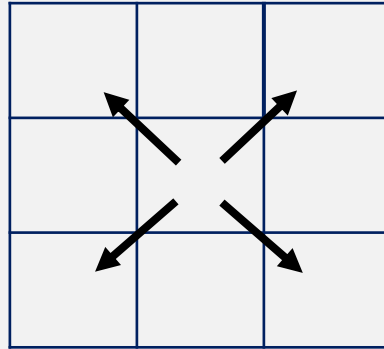
$$W = \begin{bmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{bmatrix}$$

Spatial weights matrix based on continuity can be used for **vector or raster data**.

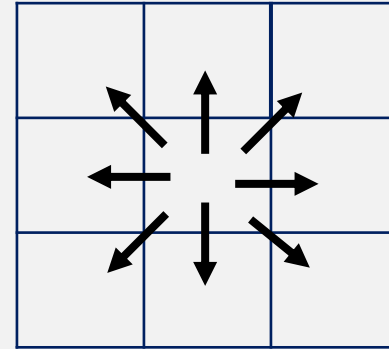
If it is used raster or lattice data, the **continuity** can be represented by the following ways:



Contiguity edges only
i.e, Rook's case



Contiguity Corners
i.e, Bishop's Case



Contiguity edges corners
i.e, Queen's Case

$$w_{ij} = \begin{cases} 1, & \text{has common edge} \\ 0, & \text{no common edge} \end{cases}$$

$$w_{ij} = \begin{cases} 1, & \text{has common vertex} \\ 0, & \text{no common vertex} \end{cases}$$

$$w_{ij} = \begin{cases} 1, & \text{has common edge or vertex} \\ 0, & \text{no common edge and vertex} \end{cases}$$

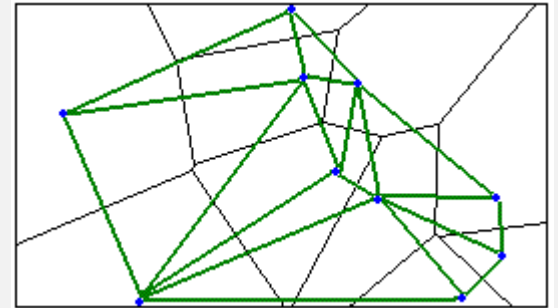
Spatial Weights Matrix: Neighbors

K nearest neighbors:

If K (the number of neighbors) is 8, the eight closest neighbors to the target feature will be included in computations for that feature. In locations where feature density is high, the spatial context of the analysis will be smaller. Similarly, in locations where feature density is sparse, the spatial context for the analysis will be larger. An advantage to this model of spatial relationships is that it ensures there will be some neighbors for every target feature, even when feature densities vary widely across the study area.

Delaunay triangulation

The Delaunay Triangulation option constructs neighbors by creating Voronoi triangles from point features or from feature centroids such that each point or centroid is a triangle node. Nodes connected by a triangle edge are considered neighbors. Using Delaunay triangulation ensures every feature will have at least one neighbor even when data includes islands or widely varying feature densities.



Spatial Weights Matrix: Geoda Tools->Weights Manager

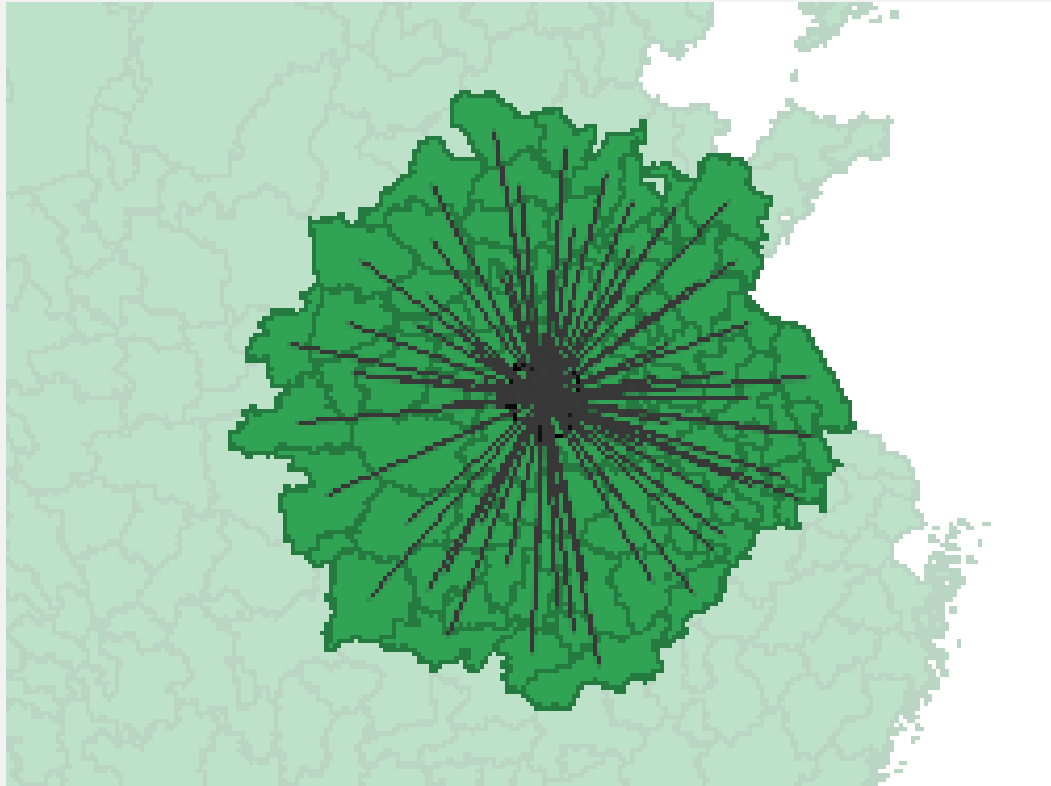


Queen's Contiguity
Contiguity edges corners



Connectivity Graph
Delaunay triangulation

Spatial Weights Matrix: Geoda Tools->Weights Manager



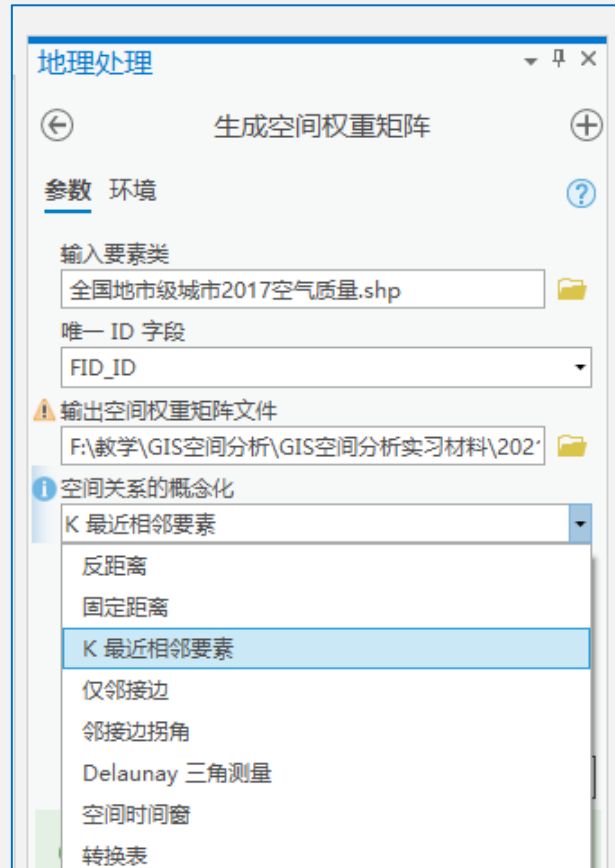
Connectivity Graph Using 500000 m as threshold distance

2. Spatial Weights Matrix

Generally, there are non-adjacent relationship between space units and themselves.

$$W = \begin{bmatrix} 0 & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & 0 \end{bmatrix}$$

“生成空间权重矩阵” Tool in ArcGIS Pro



3. Global Spatial Autocorrelation

- **Moran's I**
- **General G**
- *Join Counts*

<https://spatialanalysisonline.com/HTML/index.html>

- *Bivariate Moran's I (Geoda)*
-



Moran's I

Moran's I, which was firstly proposed by Patrick Alfred Pierce Moran in 1950, is used to measure spatial autocorrelation based on feature location and attribution values.



Moran's I

- Spatial autocorrelation is characterized by a correlation that occurs among samples that are geographically close.
 - E.g. If we find a given type of bush in an area, we are more likely to find a second specimen of that same type of bush closer rather than farther away.
- The main idea behind spatially autocorrelation data is that values are not independent over space.
- This concept is based on the first law of geography proposed by Waldo Tobler: "Everything is related to everything else, but near things are more related than distant things."
- By measuring spatial autocorrelation, we can determine how spatial patterns occur in our dataset or study area.

Moran's I

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \times \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$



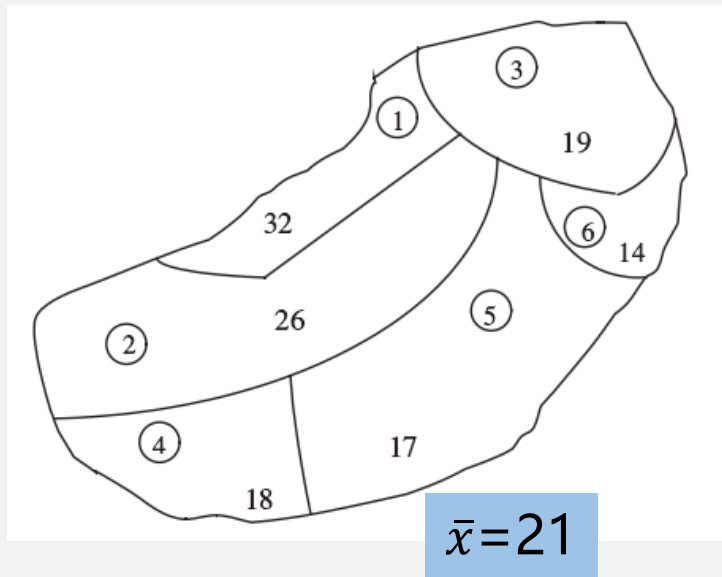
$$I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} z_i z_j}{\sum_{i=1}^n \sum_{j=1}^n w_{ij} \sum_{i=1}^n z_i^2}$$

Calculate Moran's I using standardized variables

- Where x_i , x_j are values of variable x for regions/features i and j respectively, w_{ij} is the spatial weight between regions/features i and j ; n is equal to the total number of regions/features.
- The weights w_{ij} can be defined in a number of ways
 - The most common definition is one of binary connectivity:
 - $w_{ij} = 1$, if regions i and j are contiguous,
 - $w_{ij} = 0$, otherwise

Moran's I will be in [-1, 1] if the variable values are standardized before inputting the tool.

Moran's I--Example



$$\mathbf{W} = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

The sum of the weights in the above matrix is 18

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \times \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = 0.1488$$

Moran's I—Significant Testing

- If the number of regions (features) is large, the sampling distribution of I , under the hypothesis of no spatial pattern, approaches a normal distribution, and the mean and variance of I can be used to create a Z -statistic in the usual way:

$$Z(I) = \frac{I - E[I]}{\sqrt{Var(I)}}$$

$$E[I] = \frac{-1}{n - 1}$$

$$Var(I) = E[I^2] - E[I]^2$$

$$\alpha = 0.05$$

- If $Z(I) > 1.96$, Positively Spatial autocorrelation*
- If $Z(I) < -1.96$, Negatively Spatial autocorrelation*
- If $-1.96 \leq Z(I) \leq 1.96$, not significant*

Moran's *I*—Tool and Report

Spatial Statistics Tools->Analyzing Patterns-> Spatial Autocorrelation Moran's *I*

- The ***p-value*** is a probability. For the pattern analysis tools, it is the probability that the observed spatial pattern was created by some random process. When the *p*-value is very small, it means it is very unlikely (small probability) that the observed spatial pattern is the result of random processes, so you can reject the null hypothesis. While **how small is small enough?**
- Z-scores*** are standard deviations. If, for example, a tool returns a ***z-score*** of +1.099, you would say that the result is 1.099 standard deviations. Both ***z-scores*** and ***p-values*** are associated with the standard normal distribution.

Spatial Autocorrelation Report

Moran's Index: 0.050085

z-score: 1.098550

p-value: 0.271965

Significance Level
(p-value)

0.01

0.05

0.10

0.10

0.05

0.01

Critical Value
(z-score)

< -2.58

-2.58 ~ -1.96

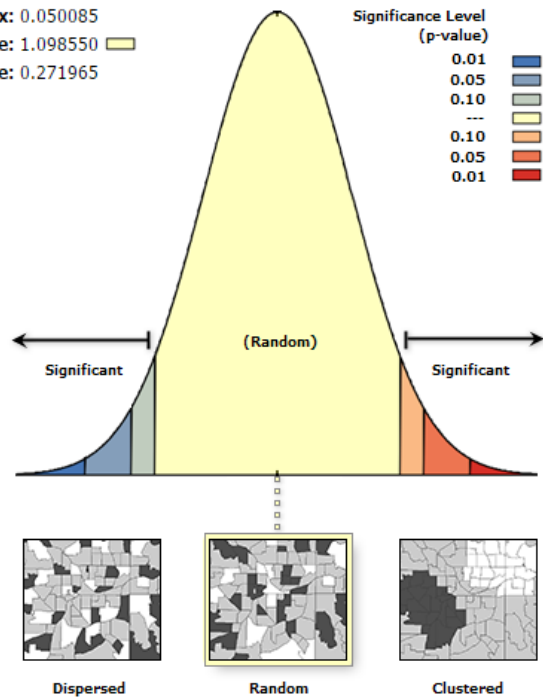
-1.96 ~ -1.65

-1.65 ~ 1.65

1.65 ~ 1.96

1.96 ~ 2.58

> 2.58



Given the z-score of 1.09854971621, the pattern does not appear to be significantly different than random.

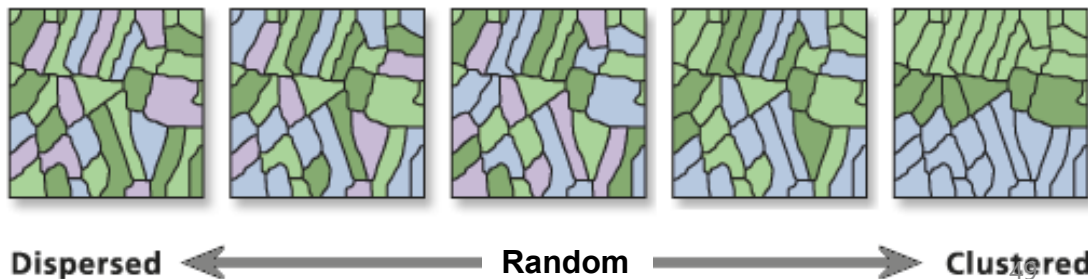
Moran's I--Example

Moran' s /



Conclusion:

- Moran' s / = 0.295
- Clustered Spatial distribution



Moran's I--Potential applications

- Help identify an appropriate neighborhood distance for a variety of spatial analysis methods by finding the distance where spatial autocorrelation is strongest.
- Measure broad trends in ethnic or racial segregation over time—is segregation increasing or decreasing?
- Summarize the diffusion of an idea, disease, or trend over space and time—is the idea, disease, or trend remaining isolated and concentrated, or spreading and becoming more diffuse?
-

3. Global Spatial Autocorrelation

- Moran's I
- General G
- *Join Count*

<https://spatialanalysisonline.com/HTML/index.html>

- *Bivariate Moran's I (Geoda)*
-



General G

- General G is a statistic which measures the degree of clustering for either high or low values.

$$G(d) = \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij}(d) x_i x_j}{\sum_{i=1}^n \sum_{j=1}^n x_i x_j}$$

Where $w_{ij}(d)$ are spatial weights.

- Under null hypothesis or random distribution, the expected mean value of General G :

$$E[G] = \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij}(d)}{n(n-1)}$$

$$Z(G) = \frac{G - E[G]}{\sqrt{\text{Var}[G]}}$$

If $G(d) > E(G)$ and Z value is significant, high-high cluster

If $G(d) < E(G)$ and Z value is significant, low-low cluster

If $G(d) \approx E(G)$, random distribution

General G

地理处理

高/低聚类 (Getis-Ord General G)

参数 环境

输入要素类

ObsData911Calls.shp

输入字段

Calls

☒ 生成报表

空间关系的概念化

反距离

距离法

欧氏

标准化

行

距离范围或距离阈值

运行

General G 观测值: 0.014020

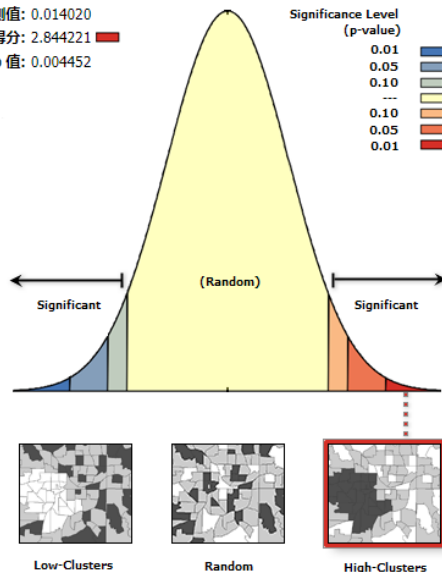
z 得分: 2.844221

p 值: 0.004452

Significance Level
(p-value)

Critical Value
(z-score)

0.01 < -2.58
0.05 -2.58 ~ -1.96
0.10 -1.96 ~ -1.65
... -1.65 ~ 1.65
0.10 1.65 ~ 1.96
0.05 1.96 ~ 2.58
0.01 > 2.58



z 得分为 2.844220771651902, 则随机产生此 高聚类 模式的可能性小于 1%。

General G 汇总

General G 观测值: 0.014020

General G 期望值: 0.011628

方差: 0.000001

z 得分: 2.844221

p 值: 0.004452

General G—Potential Applications

- Look for unexpected spikes in the number of emergency room visits, which may indicate an outbreak of a local or regional health problem.
- Compare the spatial pattern of different types of retail within a city to see which types cluster with competition to take advantage of comparison shopping (automobile dealerships, for example) and which types repel competition (fitness centers or gyms, for example).
- Summarize the level at which spatial phenomena cluster to examine changes at different times or in different locations. For example, it is known that cities and their populations cluster. Using High/Low Clustering (Getis-Ord General G), you can compare the level of population clustering within a single city over time (analysis of urban growth and density).
-

4. Local Indicators of Spatial Association (LISA)

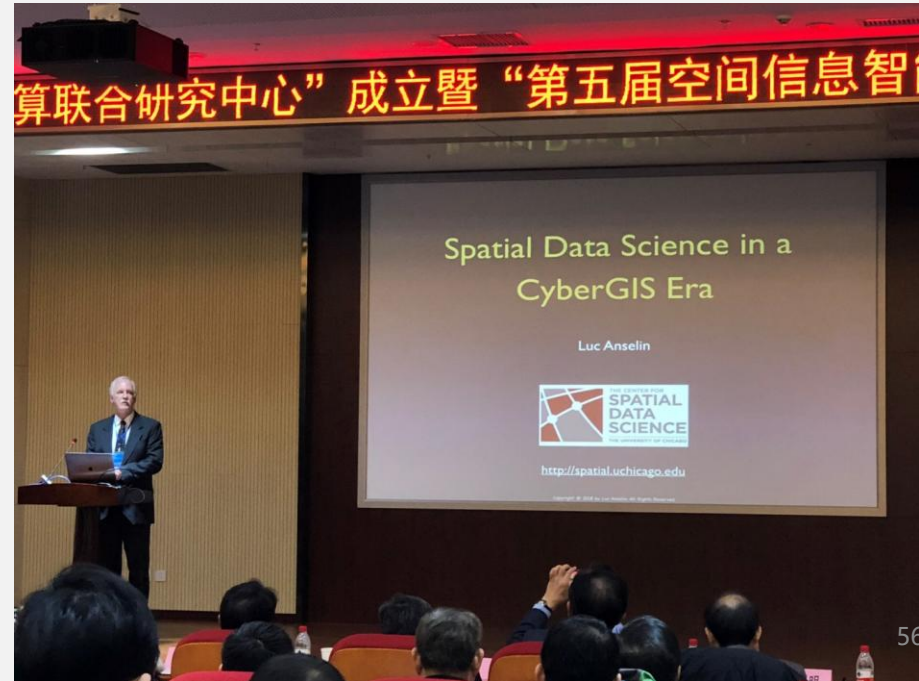
- Local Moran's I
- Getis-Ord G_i^*



Local Moran's I

Local Moran's I , which was proposed by Luc Anselin in 1995, identifies spatial clusters of features with high or low values for a given set of features (**Input Feature Class**) and an analysis field (**Input Field**).

Anselin L. Local Indicators of Spatial Association—LISA[J]. Geographical Analysis, 1995, 27(2):93-115.



Local Moran's I

Global Moran's I statistic:

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \times \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

The local Moran's I statistic:

$$I_i = \frac{x_i - \bar{x}}{S_i^2} \sum_{j \neq i} w_{ij} (x_j - \bar{x})$$

$$S_i^2 = \frac{\sum_{j \neq i} (x_j - \bar{x})^2}{n - 1}$$

- The sum of local Moran's I_i is equal to, up to a constant of proportionality, the global Moran's I : $\sum I_i = I$

Z statistic of I_i :

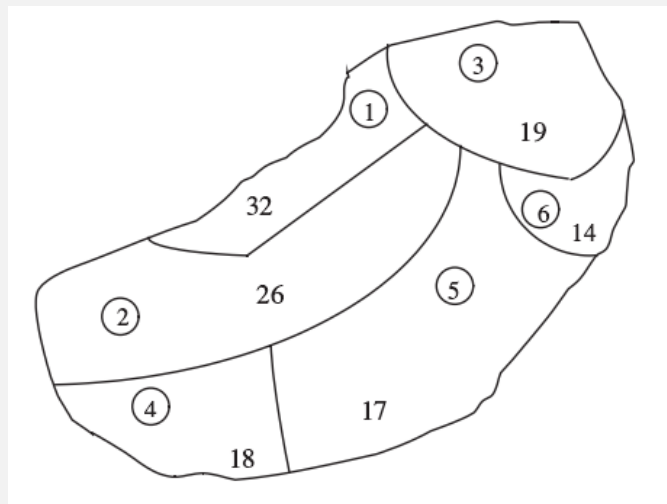
$$z_{I_i} = \frac{I_i - E[I_i]}{\sqrt{V[I_i]}}$$

Where

$$\begin{aligned} E[I_i] &= -\frac{\sum_{j=1, j \neq i}^n w_{ij}}{n - 1} \\ V[I_i] &= E[I_i^2] - E[I_i]^2 \end{aligned}$$



Local Moran's *I*--Example

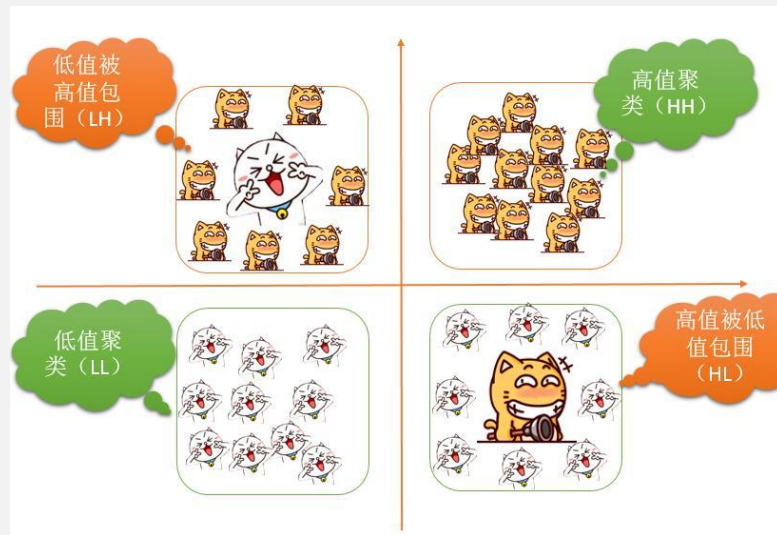


$$\bar{x} = 21,$$
$$n = 6,$$
$$S_1^2 = 20.6$$

$$I_1 = 1.602$$

$$I_i = \frac{x_i - \bar{x}}{S_i^2} \sum_{j \neq i} w_{ij} (x_j - \bar{x})$$

$$S_i^2 = \frac{\sum_{j \neq i} (x_j - \bar{x})^2}{n - 1}$$



Local Moran's I --Example

地理处理

← 聚类 and 异常值分析 (Anselin Local Moran's I) →

参数 环境

输入要素类
ObsData911Calls

输入字段
Calls

输出要素类
ObsData911Calls_ClustersOutliers

空间关系的概念化
反距离

距离法
欧氏

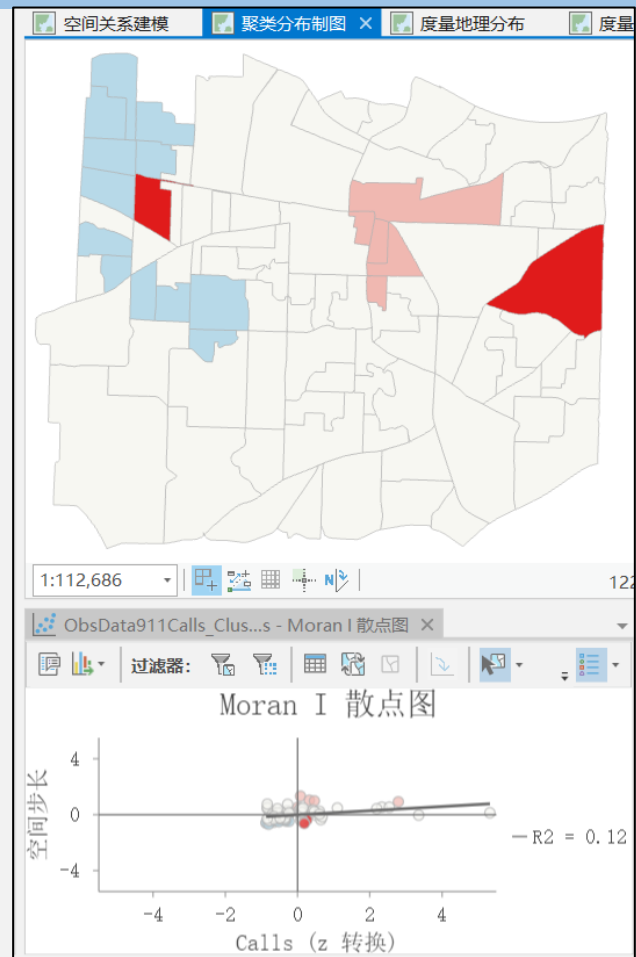
标准化
行

距离范围或距离阈值

☐ 应用错误发现率 (FDR) 校正

置换检验次数 99

运行



Local Moran's I —Potential Applications

- Applications can be found in many fields including economics, resource management, biogeography, political geography, and demographics.
 - Where are the sharpest boundaries between affluence and poverty in a study area?
 - Are there locations in a study area with anomalous spending patterns?
 - Where are the unexpectedly high rates of diabetes across the study area?
 -

4. Local Indicators of Spatial Association (LISA)

- Local Moran's I
- Getis-Ord G_i^*



Getis-Ord G_i^*

Getis and Ord G_i^* statistic using the following equation:

$$G_i^* = \frac{\sum_{j=1}^n w_{ij} x_j - \bar{x} \sum_{j=1}^n w_{ij}}{S \sqrt{\left[n \sum_{j=1}^n w_{ij}^2 - \left(\sum_{j=1}^n w_{ij} \right)^2 \right] / (n-1)}}$$

$$\bar{X} = \frac{\sum_{j=1}^n x_j}{n}$$
$$S = \sqrt{\frac{\sum_{j=1}^n x_j^2}{n} - (\bar{X})^2}$$

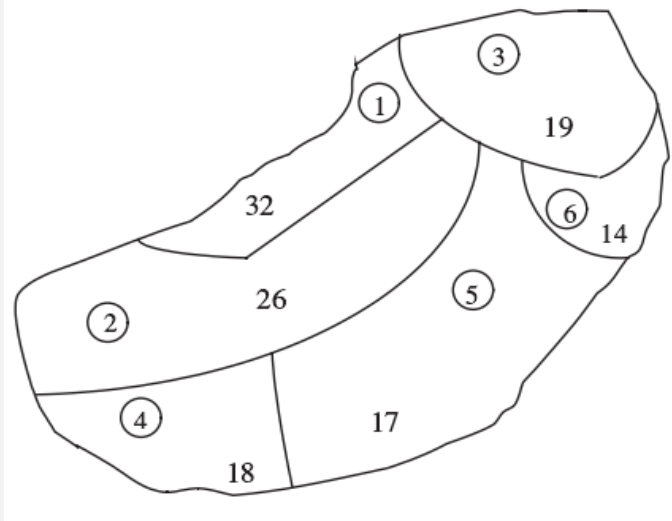
$w_{ij}(d)$ can be 0, 1 or other values.

Significant Test: Z Statistic of G_i^*

Getis A, Ord J K. *The Analysis of Spatial Association by Use of Distance Statistics*[J]. *Geographical Analysis*, 1992, 24(3):189–206.

Ord J K, Getis A. *Local Spatial Autocorrelation Statistics: Distributional Issues and an Application*[J]. *Geographical Analysis*, 1995, 27(4):286-306.

Getis-Ord G_i^* --Example



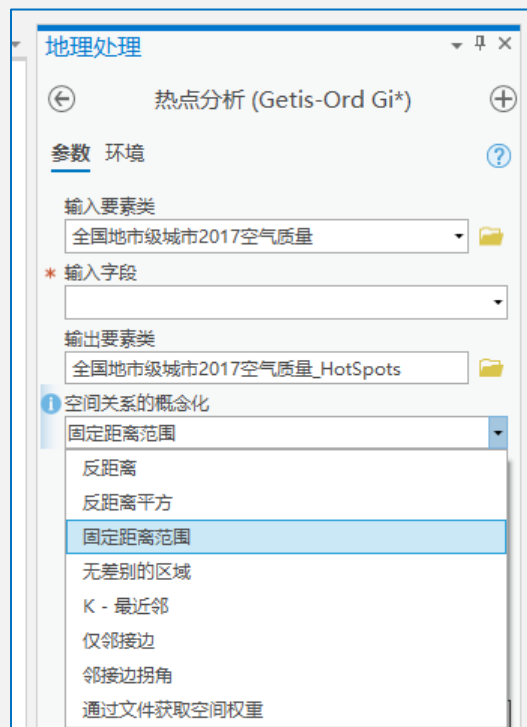
Use weights equal to 1 for regions 1, 2, and 3, and weights equal to 0 for other regions

$$G_i^* = \frac{\sum_{j=1}^n w_{i,j} X_j - \bar{X} \sum_{j=1}^n w_{i,j}}{s \sqrt{\frac{[n \sum_{j=1}^n w_{i,j}^2 - (\sum_{j=1}^n w_{i,j})^2]}{n-1}}}$$

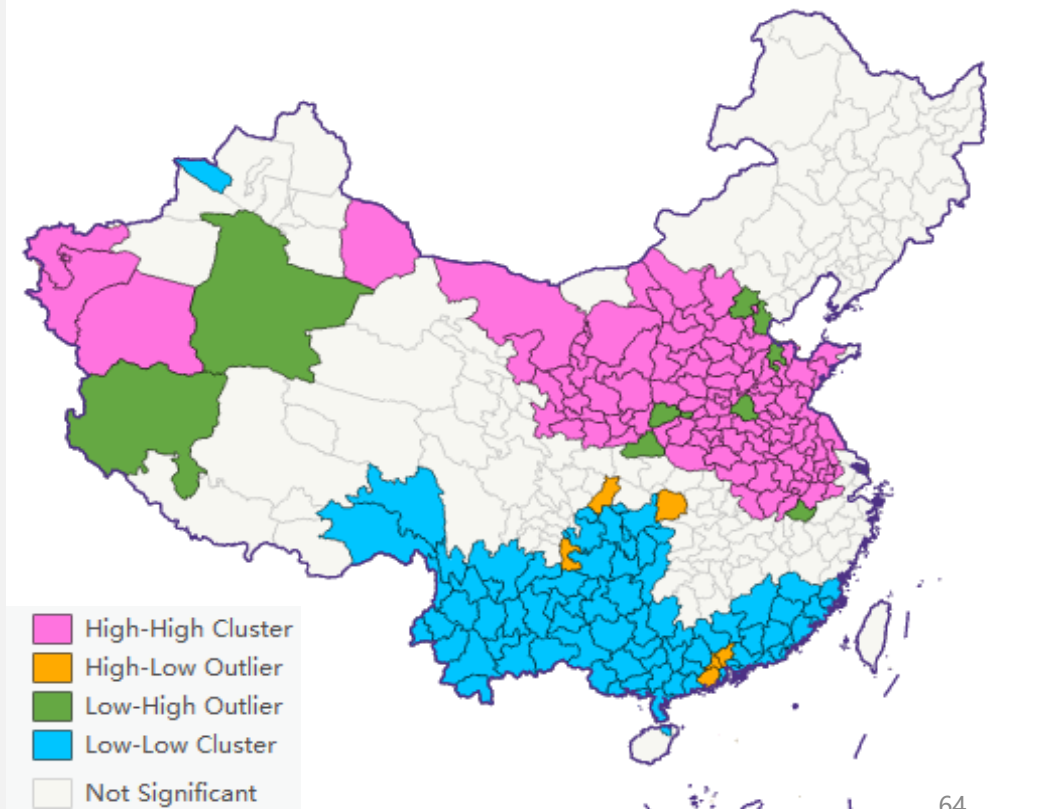
$$= \frac{77 - 3(21)}{6.69 \sqrt{(6(3) - 9)/5}} = 1.56$$

Getis-Ord G_i^* --Tool and Example

ArcGIS Pro Tool: Hot Spot Analysis (Getis-Ord G_i^*)



The spatial distribution map of Getis-Ord G_i^*



Getis-Ord G_i^* --Example

地理处理

热点分析 (Getis-Ord G_i^*)

参数 环境

输入要素类

ObsData911Calls

输入字段

Calls

输出要素类

ObsData911Calls_HotSpots

空间关系的概念化

固定距离范围

距离法

欧氏

距离范围或距离阈值

自身潜力字段

☐ 应用错误发现率 (FDR) 校正

运行

内容



搜索



绘制顺序

聚类分布制图

ObsData911Calls_HotSpots

Gi_Bin

- 置信度为 99% 的冷点
- 置信度为 95% 的冷点
- 置信度为 90% 的冷点
- 不具有显著性
- 置信度为 90% 的热点
- 置信度为 95% 的热点
- 置信度为 99% 的热点

图表

Calls 的直方图

相似性搜索

构建平衡区域

优化的热点分析

热点分析

优化的异常值分析

空间异常值检测

聚类和异常值分析

基于密度的聚类

空间约束聚类

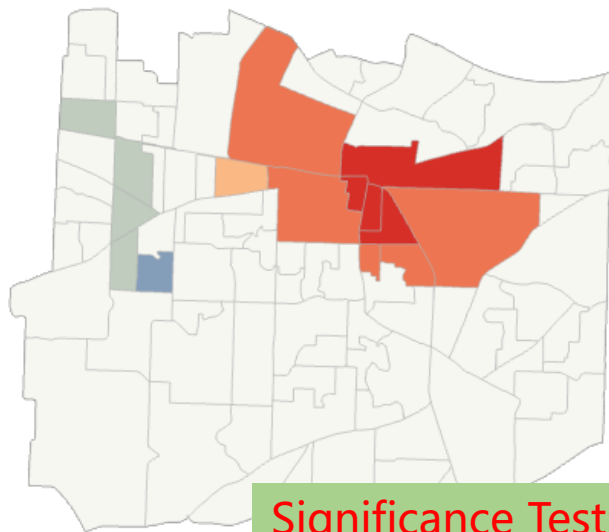
聚类分布制图

度量地理分布

度量地理分布之线性方向

分析模式

空间关系建模之PM25



Significance Test: Each Feature has its Z -Score, p -value and number of neighbors of G_i^* .

1:112,686

ObsData911Calls_HotSpots

字段: 添加 计算

选择: 按属性选择

缩放至

切换

清除

删除

复制

	OBJECTID *	Calls	GiZScore Fixed 6331	GiPValue Fixed 6331	NNeighbors Fixed 6331	Gi_Bin Fixed 6331
1	47	27	4.208725	0.000026	11	3
2	48	34	4.602351	0.000004	13	3
3	57	38	4.633914	0.000004	13	3
4	66	104	3.256198	0.001129	9	3

Getis-Ord G_i^* —Potential Applications

- Applications can be found in crime analysis, epidemiology, voting pattern analysis, economic geography, retail analysis, traffic incident analysis, and demographics. Some examples include the following:
 - Where is the disease outbreak concentrated?
 - Where are kitchen fires a larger than expected proportion of all residential fires?
 - Where should the evacuation sites be located?
 - Where/When do peak intensities occur?
 - Which locations and at during what time periods should we allocate more of our resources?
 -



Lect. 5 Spatial Correlation Analysis

- **Traditional Correlation Analysis in Geography**
 1. Person's Correlation Coefficient
 2. Spearman's Rank Correlation Coefficient
 3. Partial Correlation Coefficient
 4. Multiple Correlation Coefficient
- **Spatial Autocorrelation Analysis**
 1. The Concept of Spatial Autocorrelation
 2. Spatial Weights Matrix
 3. Global Spatial Autocorrelation
 4. Local Indicators of Spatial Association
- **Study Cases**



Study Cases

- Spatiotemporal influencing factors of energy efficiency in 43 european countries: A spatial econometric analysis-2024
- Spatial and temporal distribution characteristics of haze and pollution particles in China based on spatial statistics-2022
- Spatial equity analysis of urban public services for electric vehicle charging-Implications of Chinese cities-2022
- Landscape ecological risk assessment and its driving factors of multi-mountainous city-2023



References

1. Chapter 5 & 8 in Rogerson' s 《Statistical methods for geography》 .
2. Section 2.5 in 《Geospatial Analysis 6th Edition, 2020 update》
<https://spatialanalysisonline.com/HTML/index.html>.
3. Anselin L. Local Indicators of Spatial Association—LISA[J]. Geographical Analysis, 1995, 27(2):93-115.
4. Related papers.

Reading and Assignments

1. Read the references.
2. What is the basic idea of spatial autocorrelation? What are the typical applications of spatial autocorrelation?
3. Do exercises 1-3 (pages 102 and 103 in textbook of ***Statistical Methods for Geography***) .
4. Do Lab 5.

The report of Lab 5 should be submitted before
20:00, April 4.

